



# Development Guideline for ABAP

## History

| Version | Status   | Date           | Author(s)                    | Explanation                                                                                                                                                                      |
|---------|----------|----------------|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.0     | approved | 2006/01/11     | K. Schulze                   | Conversion of version 2.0 into an appendix to guideline RL_009_Softwareentwicklungsmethodik_001 (software development methodology)                                               |
| 2.0     | approved | 2007/06/30     | K. Schulze                   | Reworking of form and content                                                                                                                                                    |
| 3.0     | approved | 2008/01/08     | K. Schulze                   | Changes due to the harmonization of the guidelines within the E.ON IS Group<br>- Naming conventions<br>- Data definitions<br>- Function module parameters<br>- Formal parameters |
| 4.0     | approved | 2009/09/25     | K. Schneider                 | Change:<br>- Structure of document<br>- Creation of checklist<br>- Addition of jump marks<br>- Adjustment to accord with new SAP technologies<br>- Completion of document        |
|         |          |                | H. Thurm                     | Finalization                                                                                                                                                                     |
| 4.1     | approved | 2011/04/26     | H. Schönenberg<br>H. Thurm   | Change:<br>Addition of recording DB-Changes (Chapter 3.31.5)<br>Finalization                                                                                                     |
| 5.0     | approved | 2012 June      | R. Várnagy,<br>G. Váradi     | Reworked structure and content                                                                                                                                                   |
| 5.1     | approved | 2014 June      | H. Schönenberg<br>G. Váradi  | New rules because of QA audit                                                                                                                                                    |
| 6.0     | approved | 2015 July      | P. Jecke<br>G. Váradi        | Restructuring and several additions                                                                                                                                              |
| 6.1     | approved | 2015 October   | G. Váradi                    | New rules introduced for HANA in the rulebook                                                                                                                                    |
| 6.2     | approved | 2016 September | G. Váradi                    | smaller changes                                                                                                                                                                  |
| 7.0     | In work  | 2017 March     | M.Nitschke,<br>O. Luchtefeld | Virtual Forge Code Profiler Changes                                                                                                                                              |

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# 1 Introduction

## 1.1 Objectives

This document determines requirements for performant, secure, robust and maintainable SAP code implementation. This guideline sets rules regarding:

- developing customer objects
- copy of SAP standard objects
- modify SAP standard objects
- import code developed by 3<sup>rd</sup> Party

These guidelines are valid for all development activities performed in the SAP-Systems at EON by internal or external employees. The objective of these rules and recommendations in this document is to ensure correct functionality as well as efficient and thus cost-effective maintenance for all customer-owned or modified objects in EBS's SAP systems.

Standardized ABAP coding is an essential factor for success in an efficient processing of support requests in the various systems for various application areas and customers. Furthermore there are legal and contractual requirements to be obeyed (see chapter 3.7 & 3.8).

In addition security has high criticality in SAP environments. Connections into the web and the opening for Digital Transformation give space for scenarios to penetrate a system from outside of the Corporate Network.

The usage of an automated and standardized Code quality check is necessary to fulfill the different rules and recommendations from standard organizations. E.ON is using the Code Profiler from the leading SAP Security Company Virtual Forge. The Code Profiler covers several industry standards with more than 300 test cases.

All additional E.ON requirements are implemented with customer test cases, prescribed by E.ON. The Code Profiler supports currently [22 SAP landscapes](#) (Connect -> Community "SAP Basis & ABAP Objects"-> Code Profiler Blog). In all other systems the quality check will be done manually based on the code profiler rules.

A documentation of the functionality of the Code Profiler and of all rules (realized with test cases) can be found here ( Standard documentation of Code Profiler):

The main documents can be found [here](#)

- **Code Profiler Testcases** with explanation of each test case created by E.ON and Virtual Forge.
- E.ON and Virtual Forge Dokumentations how to handle the CodeProfiler.
- Style Guide Application Security

Guidelines, patterns and more see: Chapter 3

## 1.2 Findings

In July / August 2017 identified findings after the initial code profiler scans won't be regarded for the time being. Only findings found starting from October 2017 are required to be fixed, ditto for systems without code profiler installed

A "get clean" project must be set up to fix the high number of findings in legacy code. This is in discussion and therefore not in scope of this guideline.

## 2 Procedures

### 2.1 Change and copy customer objects

#### 2.1.1 Conditions / prerequisites

- New object's name must fit the naming conventions
- Change history must contain the old objects name and the source production system
- It's recommended to fix findings in former developed code wherever this is apparent. Only then E.ON is able to improve code quality step by step.
- Break points are not allowed, DB update without Lock neither.
- Reports must have a transaction and an authority check.
- The system ID of the original system of copied customer code must be changed to the new system ID. This copied customer code is regarded as new development and must obey the rules. Otherwise non-compliant code is being multiplied and impairs quality instead of improving quality.
- Copied objects are regarded as new objects. All findings need to be fixed, also the legacy findings from the copied version.

### 2.2 Copy live SAP standard code

#### 2.2.1 Conditions / prerequisites

Every copy of SAP standard code (so called clone) is a challenge in case of patching, upgrading or a release change. It's expensive and error prone to synchronize these clones manually with the patched SAP code. While handling modifications and enhancements is a standard step in a system upgrade, custom clones can't even be identified and might show up as syntax or semantic errors later.

The criteria:

- The copy must be marked as such with a comment in the function description!
- Reason why copying instead of standard modification is required
- Don't change neither the original transaction / object name nor the type but precede each transaction / object name a Z or the namespace
- Unused original SAP code must be deleted.
- Copied objects are regarded as new objects. All findings need to be fixed, also the legacy findings from the copied version.
- SAP standard documentation must be enhanced with the change Information.
- Generated source must not be modified. It's allowed to be copied, but it must be handled like a standard development. A change history, transaction and authority check is mandatory.
- Customer objects created because of OSS notes must follow this guideline. The following adjustments are mandatory:

- Implement an authority check if it is a report.
- Lock DB-Tables before DB changes.
- Create a description and change history (include the OSS Note number).

## 2.3 Modify SAP standard code

### 2.3.1 Conditions / prerequisites

Always give preference to enhancements, exits or modifications before considering copying SAP standard!

In exceptional cases, it may be required to copy SAP standard programs into the customer namespace and modify them. But copying standard code also has the disadvantage of breaking this application's link to SAP's development and maintenance path and of changes not being incorporated into the standard. You should therefore carefully balance the advantages and disadvantages. Obey the following basic rules:

The criteria:

- Use enhancements whenever feasible and limit the amount of modification.
- Always use the modification assistant.
- Modifications must be marked with BEGIN-END. Obsolete Coding must be deleted.

```
*>>> BEGIN <Date> <User ID> <Reason>
*>>> END <Date> <User ID> <Reason>
eg.:
*>>> BEGIN 01.01.2012 N4653 INC00004564645
      IF ever-ausdat EQ '99991231'.
*>>> END 01.01.2012 N4653 INC00004564645
```

- Control customer-specific functionalities must use a customizing table.
- Encapsulate your own source code in modularization units instead of writing own code in SAP source code (e.g. calling customer function modules in program source code, calling customer sub-screens for extra dynpro fields).
- Interfaces used for encapsulation should be as narrow as possible.

## 2.4 Import code developed by 3rd Party

### 2.4.1 Conditions / prerequisites

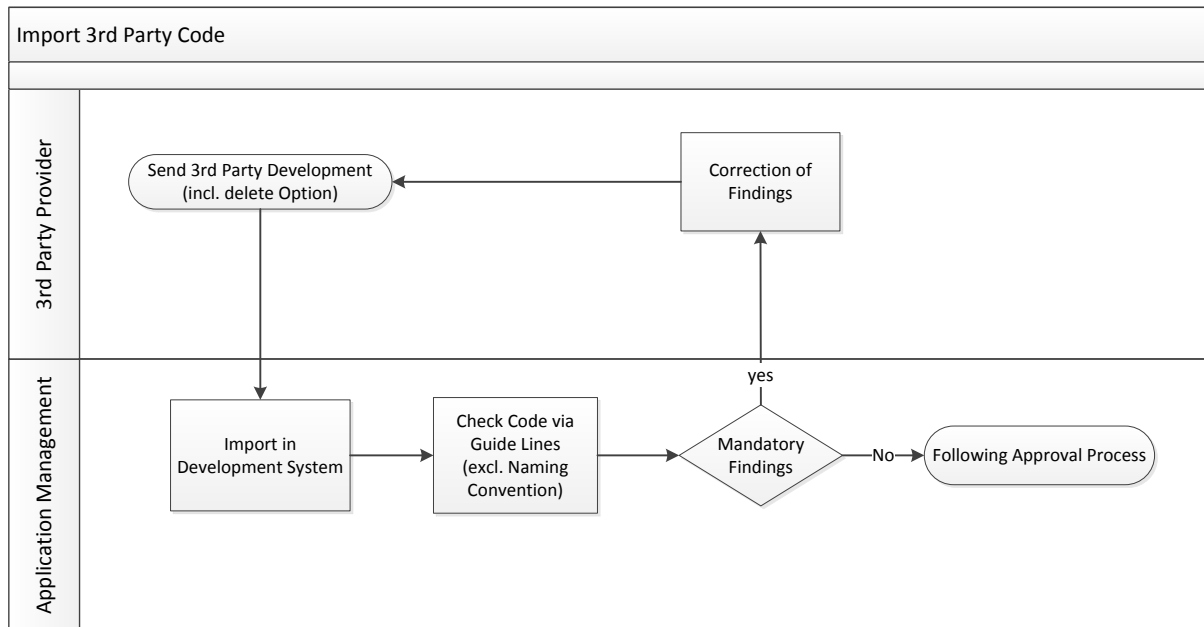
These criteria apply to:

- Code developed by 3<sup>rd</sup> party to be deployed on a production SAP system at E.ON.

The criteria:

- Code developed by 3<sup>rd</sup> party must pass the code profiler check at least the categories security, data loss prevention and compliance. If these categories are hurt but the code will still be deployed then a risk assessment must take place, triggered by the application responsible manager.
- All code fragments (e.g. BAPIs, reports, etc.) bundled in one namespace are regarded as one registered Customer Namespace.
- The provision by the supplier of 3<sup>rd</sup> party code must be appropriate (e. g. with signed email).

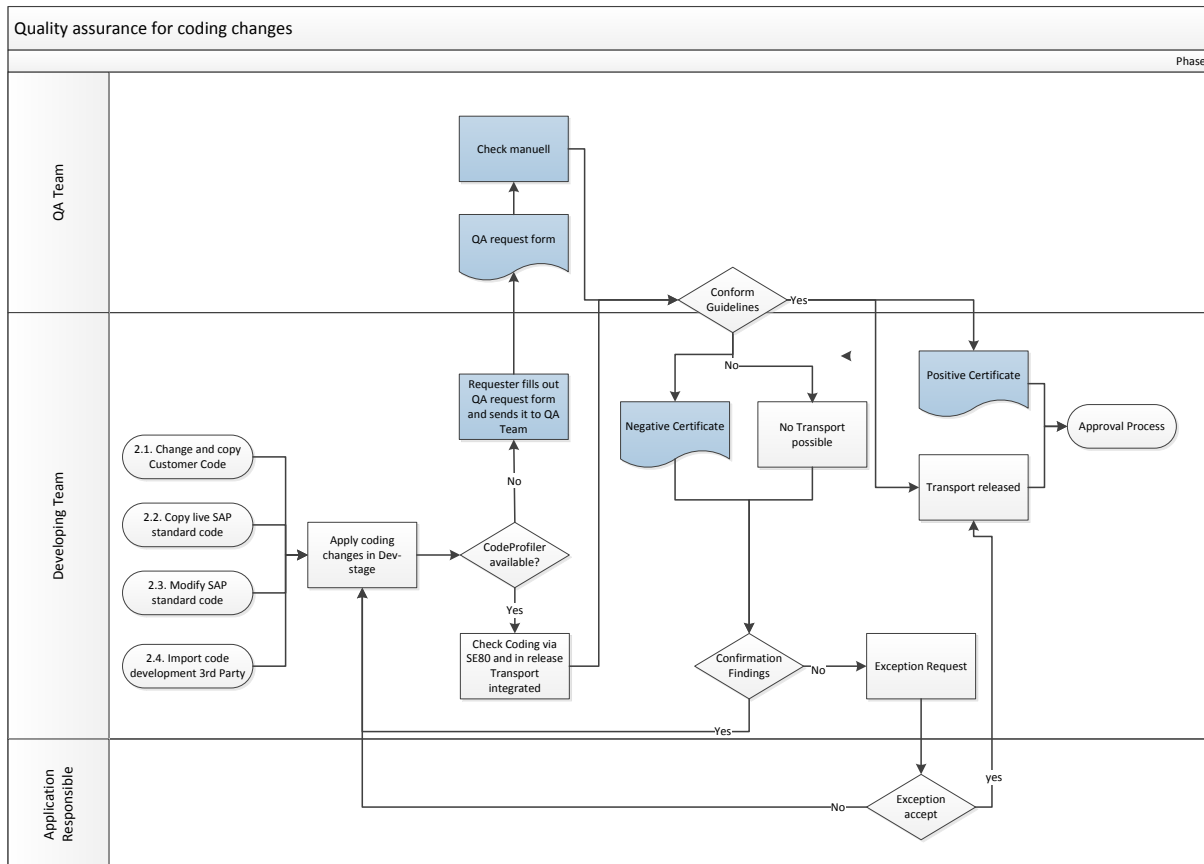
## 2.4.2 Process flow



Another option is to check the 3<sup>rd</sup> party code via Cloud Virtual Forge CodeProfiler. Its advantage is that an import into the E.ON Environment is not necessary. Furthermore the supplier can certify his code by Virtual Forge or the TÜV.

In case of a transport or installation package into a Dev-System of E.ON the supplier must provide an option to delete / cancel the whole transport / installation. This is to ensure that no 3<sup>rd</sup> party code is in the system in case of a failed quality check.





manuell

Manuel process

## 3 Appendices

### 3.1 Customer name space, modules and module codes

The namespace has to follow this format:

<Namespace>:= (Customer Namespace **Z** | Partner Namespace /<Characters>/ )

eg.: /SAPCE/, /GBE/...

**The module code in the object name must fit to the module code of its package!** (it must have the same module code).

You find all of the allowed module codes in the system you are working with in transaction SE81. You can use only those modules which you can find in the system!

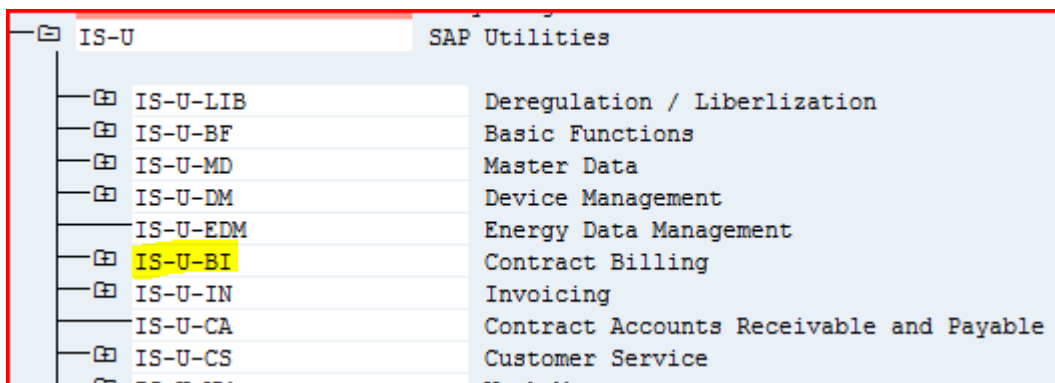
By concatenating the module code you have to use the 2 character long module code (e.g. 'CO').

There are two exceptions: the ISU and the CRM modules. Instead of ISU, we only use the letter E, and instead of CRM we use C.

If it is needed, the developer can use a more detailed name by using sub modules.

A module code can't be longer than 4 characters.

For example: ISU-Billing: EBI



| Package  | Description                              |
|----------|------------------------------------------|
| IS-U     | SAP Utilities                            |
| IS-U-LIB | Deregulation / Liberalization            |
| IS-U-BF  | Basic Functions                          |
| IS-U-MD  | Master Data                              |
| IS-U-DM  | Device Management                        |
| IS-U-EDM | Energy Data Management                   |
| IS-U-BI  | Contract Billing                         |
| IS-U-IN  | Invoicing                                |
| IS-U-CA  | Contract Accounts Receivable and Payable |
| IS-U-CS  | Customer Service                         |

ISU Function Module: Z\_EBI\_FUNCTIONS

CRM Report: ZC\_REPORT\_2015 or ZCBF\_BASIC\_FUNCTIONS

If you would like to use a different module code or have any question, [contact the QA Team](#)

## 3.2 Source code format conventions

### Coding style

- Delete **obsolete** parts of the code. Don't comment it out
- Attributes, methods, events and types must have a meaningful **English** name
- Classical Dynpro fields require **global** data objects in the function group. Keep those globals at an absolute minimum. Declare a dedicated structure for every screen - do not 'share' one global object with different screens.
- The software architecture should follow the MVC-principle: Do not mix coding that manages screen flow or appearance (as well as event handlers for screen elements) with screen-independent or business logic.
- Use always Pretty Printer
- Use local classes instead of using form routines.

### Inline comments

- All comments must be in **English**.
- Description of all objects with report source must be filled in the header.

### Documentations

- In case of SAP standard code has been copied, customer changes must be commented and specific function module documented
- It is possible to set the Short text as documentation for a data element (documentation status) or alternatively provide a proper documentation.
- A meaningful brief description must be created for each Dynpro.

### Usability

- The SPRO path must be updated along with the brief description of the function. When linking the created Z-Function module in IMG, please enter a proper short description.
- Avoid list reports. Use SALV where available.
- To each field (parameter or select options) that appears in the selection screen, a corresponding identification should be assigned in the text elements.
- The look & feel of screen dialogs must comply with the SAP standard design.
- Use a standard GUI status or create a new one for every program. A GUI title is also obligatory.
- At the end of a process there must be a feedback for the user about the application state. It can be a LOG, an ALV list or a success/error message.
- Ensure that the user always finds a way to cancel processing and leave the dynpro quickly (e.g. the user shouldn't be forced to fill all fields correctly to finally press the 'Cancel'-Button).
- Using classical dynpro screens usually does not offer SAP standard access to the user documentation like in reports. In this case create a documentation in SE61 and display it with help of a button

### Further Recommendations

- Internal tables with high number of rows should be cleared after usage.
- A table must be sorted before a READ TABLE BINARY SEARCH.

- "ORDER BY or sorting is necessary if the internal table is used in:
  - READ
  - CONCATENATE ... LINES OF itab ...
  - DELETE ... FROM/TO or INDEX
  - FIND FIRST OCCURRENCE ... IN TABLE
  - LOOP AT itab.
  - · AT ... ENDAT.
  - · EXIT/RETURN/LEAVE ...
  - · FROM/TO
  - MODIFY ... INDEX
  - PROVIDE ...
  - READ TABLE ... INDEX ...
  - REPLACE FIRST OCCURRENCE ... IN TABLE
  - SEARCH ... IN"
- Use whitelists or RegEx for external data if possible. At times, it is easy to limit user input to a limited set of valid values in a whitelist. Methods CHECK\_WHITELIST\_STR or CHECK\_WHITELIST\_TAB of class CL\_ABAP\_DYN\_PRG may be used for this purpose.
- "Cooperative response to SAP locks: Before any changes are made to the database, the object to be modified in the LUW must be locked by means of the logical locking mechanism such that concurrent access by another or the same program is rendered impossible. In other words, when changing SAP standard tables, the correct lock object should be determined first and used accordingly. In exceptional cases (e.g. XPRA) you can do without the lock object because concurrent access will be prevented in another way. The exceptions must explained (documentation included in program code) why locks are not necessary
- It is necessary to fill the secondary tables at the lock object if you want to change the data within logical connected tables, in this case it is not enough to lock only the header table.
- Directly changing global data from a form routine without parameters is forbidden (CHANGING). Exceptions must be described in the documentation of the form routine; variables must be listed in the header of the form routine.
- Generated user exits need to retain the interface.
- Includes where the Form routines are, must have the same module name as the FORM.
- Transport to all relevant clients, exception for references
- Implementation of subroutines in function modules are forbidden. The implementation must be placed in the proper include (e.g. ...F01).
- Customer (customizing) tables and views are client-dependent. First column must be the MANDT or CLIENT as the field and MANDT as the data element.
- Customizing Tables the "LOG data changes" checkbox must be checked
- "Display/Maintenance allowed" is only allowed in connection with a generated table maintenance view. For customizing table the standard record entry routine must be also selected.

### 3.3 Naming conventions for repository objects

Naming conventions significantly enhance the readability of programs and therefore the quick understanding of developments.

In addition to the fixed conventions with respect to the prefix of a development object, the following general rules apply:

- Use meaningful and descriptive **English names**.
- Make use of the **maximum** allowable length of names.
- Use a maximum of 15 characters for the name of **tables** (because of the lock object name).
- Compound names must be separated by an **underscore**.
- If there is a standard **SAP or otherwise conventional** name, give preference to that name instead of your own invention (Use SAP transaction 'SAPTERM' to find out the common naming/ description where possible.)
- Use the following characters only: 'A-Z', '0-9' and '\_' (no hyphens).
- Names **must not begin** with a number (0..9).
- The names of normal development objects in transportable development classes **must not end in "\_D"**. Objects with this suffix are reserved for temporary backups of deleted objects.
- When using **tools and wizards** (e.g. Easy Enhancement Workbench), take naming conventions into account as much as possible
- All field labels must be filled by adequate descriptions. Length of field label is recommended to be set to the maximum character length: short = 10, medium = 20, long = 40, Heading= 55.
- If it's necessary you can set the short text as documentation for a data element (documentation status) or alternatively provide a proper documentation.
- Application Component must be filled
- If you have too many objects for one package then use subpackages for modularisation.
- For a customizing table a table maintenance generator must be created either directly or through a maintenance view, and set the maintenance to allowed, where you create it.

| Object                                              | Rule                                                                                                                                                                                                               | Example                                                |
|-----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| Authorization object                                | Z<Characters>                                                                                                                                                                                                      | ZAUTH1                                                 |
| BAdi implementation class                           | <Namespace> <b>CL_IM</b> <Module>_<Characters>                                                                                                                                                                     | /SAPCE/CL_IM_TR_EVD                                    |
| BAPI Function module                                | <Namespace> <b>BAPI</b> <Business object>_<Method>                                                                                                                                                                 | ZBAPI_ZBUSINST_GET_STATUS                              |
| Business object                                     | <Namespace> <b>BUS</b> <Characters>                                                                                                                                                                                | ZBUS_ISTALLATION                                       |
| Class                                               | <Namespace> <b>CL</b> <Module>_<Characters>                                                                                                                                                                        | ZCL_PM_MAT                                             |
| Class Exception classes                             | <Namespace> <b>CX</b> <Module>_<Characters>                                                                                                                                                                        | ZCX_PM_EXC01                                           |
| Class Interface                                     | <Namespace> <b>IF</b> <Module>_<Characters>                                                                                                                                                                        | ZIF_PM_BASE                                            |
| CMOD                                                | <Namespace><Characters>                                                                                                                                                                                            | ZPREBILL                                               |
| Domain                                              | <Namespace><Module> <b>D</b> <Characters>                                                                                                                                                                          | ZCMD_D_PART_NAME                                       |
| Data element                                        | <Namespace><Module> <b>D</b> <Characters>                                                                                                                                                                          | ZEBID_CONTRACT                                         |
| Enhancement implementation                          | <Namespace><Module>_<Characters>_I<2Digits>                                                                                                                                                                        | /GBE/MM_ENH_I01                                        |
| Enhancement implementation include                  | <Enhancement implementation>_<2Digits>                                                                                                                                                                             | /GBE/MM_ENH_I01_05                                     |
| Enhancement Point                                   | <Namespace><Module>_<Characters>_P<2Digits>                                                                                                                                                                        | ZETO_EP_P02                                            |
| Enhancement Section                                 | <Namespace><Module>_<Characters>_SC<2Digits>                                                                                                                                                                       | ZETO_EP_SC02                                           |
| Enhancement Spot                                    | <Namespace><Module>_<Characters>_S<2Digits>                                                                                                                                                                        | ZETO_EP_S02                                            |
| Form: Adobe forms                                   | <Namespace><Module> <b>AF</b> <Characters>                                                                                                                                                                         | ZCAAF_FORM1                                            |
| Form: Adobe forms interface                         | <Namespace><Module> <b>AIF</b> <Characters>                                                                                                                                                                        | ZCAAIF_FORM1                                           |
| Form: SAPscript Form class                          | <Namespace><Module> <b>FC</b> <Characters>                                                                                                                                                                         | ZEBIFC_BILL                                            |
| Form: SAPscript Form name                           | <Namespace><Module> <b>SS</b> <Characters>                                                                                                                                                                         | ZEBISS_BILL                                            |
| Form: SAPscript Standard text                       | <Namespace><Module> <b>ST</b> <Characters>                                                                                                                                                                         | ZEBIST_HEADER                                          |
| Form: SAPscript Style name                          | <Namespace><Module> <b>SN</b> <Characters>                                                                                                                                                                         | ZEBISN_BIG                                             |
| Form: Smart forms                                   | <Namespace><Module> <b>SF</b> <Characters>                                                                                                                                                                         | ZMMSF_QUANTITY                                         |
| Form: Smartforms Style name                         | <Namespace><Module> <b>SSN</b> <Characters>                                                                                                                                                                        | ZEBISSN_BIG                                            |
| Form: Smartforms text module                        | <Namespace><Module> <b>STM</b> <Characters>                                                                                                                                                                        | ZEBISTM_HEADER                                         |
| Function group Class include                        | L<Function group> <b>C</b> <2Digits>                                                                                                                                                                               | LZEEDM_PROFE02                                         |
| Function group Event include                        | L<Function group> <b>E</b> <2Digits>                                                                                                                                                                               | LZEEDM_PROFE02                                         |
| Function group Form routines include                | L<Function group> <b>F</b> <2Digits>                                                                                                                                                                               | LZEEDM_PROFF01                                         |
| Function group Input module include                 | L<Function group>I<2Digits>                                                                                                                                                                                        | LZEEDM_PROFI01                                         |
| Function group Output module include                | L<Function group> <b>O</b> <2Digits>                                                                                                                                                                               | LZEEDM_PROFO01                                         |
| Function group TOP include                          | L<Function group> <b>TOP</b>                                                                                                                                                                                       | LZEEDM_PROFTOP                                         |
| Function Module                                     | <Namespace>_<Module>_<Characters>                                                                                                                                                                                  | Z_EBI_854_STATUS_SET                                   |
| Lock object                                         | By Z tables: E<Z table name><br>By Partner Namespace: /partnernamespace/E<remainder Characters of Table name><br>By SAP Tables: EZ<SAP table name>                                                                 |                                                        |
| Method                                              | <Verb>_<Characters><br><Verb> := <b>get, set, calculate, etc.</b><br><b>Event-Methods:</b> ON_<Characters>                                                                                                         | SET_STATUS                                             |
| Message class                                       | <Package>_<optional characters>                                                                                                                                                                                    | ZEDM_PORTF_001                                         |
| Query                                               | <Namespace><Module> <b>Q</b> <Characters>                                                                                                                                                                          | ZBCQ_TESTQUERY                                         |
| Query - User group                                  | <Namespace><Module> <b>UG</b> <Characters>                                                                                                                                                                         | ZBCUG_GROUP1                                           |
| Query – InfoSet                                     | <Namespace><Module> <b>IS</b> <Characters>                                                                                                                                                                         | ZBCIS_TESTINFOSET                                      |
| Search help                                         | <Namespace><Module> <b>H</b> <Characters>                                                                                                                                                                          | ZEMDH_CONTRACT                                         |
| Structure                                           | <Namespace><Module> <b>S</b> <Characters>                                                                                                                                                                          | /SAPCE/EEDMS_PROFVAL                                   |
| Structure Append structure                          | <Structure/Table>_A<Number><br>by standard tables: Z<module>S_<sap.table.name>_A<Number><br>or if it's possible Cl_<Structure/Table><br>The technical names for new components used in appends must start with ZZ. | ZEBIT_INST_A1<br>ZEMDS_EVER_A1<br>CI_EANL<br>ZZ_ANLAGE |
| Table: Customizing table                            | <Namespace><Module> <b>C</b> <Characters>                                                                                                                                                                          | /GBE/MMC_MATCUST                                       |
| Table: other tables (e.g.: Application Table)       | <Namespace><Module> <b>T</b> <Characters>                                                                                                                                                                          | ZEDMT_TEMP_GERNR                                       |
| Table: Index                                        | Z<Digit> for SAP standard tables, otherwise free to decide                                                                                                                                                         | Z1                                                     |
| Table type                                          | <Namespace><Module> <b>B</b> <Character>                                                                                                                                                                           | ZPSB_HIST                                              |
| Type group                                          | The object name may be selected freely. Type groups are obsolete                                                                                                                                                   |                                                        |
| View                                                | <Namespace><Module> <b>V</b> <Characters>                                                                                                                                                                          | ZMMV_ALLMAT                                            |
| Viewcluster                                         | <Namespace><Module> <b>VC</b> <Characters>                                                                                                                                                                         | ZPSVC_IIFD_DATA                                        |
| Workflow                                            | See document "Development for SAP Business Workflow"                                                                                                                                                               |                                                        |
| Separately not listed objects (e.g.: Reports, etc.) | <Namespace><Module>_<Characters>                                                                                                                                                                                   |                                                        |

### 3.4 Naming conventions for data declarations

- **Global Data:** it is “visible” everywhere in the main object.
- **Local Data:** it is only “visible” in the form routines, methods or in the function modules where you declared it, nowhere else.
- **Attributes, types and aliases** of classes must be named as global data/constants/types.
- If you create an object which is called through SAP customizing, it **must have the same interface** like the sample. (e.g.: FI-CA Event sample function modules).
- Use the following characters only: 'A-Z', '0-9' and '\_' (no hyphens)

#### Every data objects must have the pattern:

<Data object> := <Scope><Type>\_<Characters>

<scope> := (G | L | I | E | C | R | T)

<type> := (C | V | S | T | R)

Every scope letter is attached to a meaning according to the table below:

|           |                                                     |
|-----------|-----------------------------------------------------|
| <b>G:</b> | global object                                       |
| <b>L:</b> | local object                                        |
| <b>I:</b> | using/importing parameter                           |
| <b>E:</b> | exporting parameter (not allowed for Form routines) |
| <b>C:</b> | changing parameter                                  |
| <b>R:</b> | returning parameter (not allowed for Form routines) |
| <b>T:</b> | types                                               |

Every 'type' letter is attached to a meaning according to the table below:

|           |                        |
|-----------|------------------------|
| <b>C:</b> | constant               |
| <b>V:</b> | general data field     |
| <b>S:</b> | structure or work area |
| <b>T:</b> | internal table         |
| <b>R:</b> | object reference       |

The following naming conventions are defined for parameters and select options:

select option

<Data object> := S\_<Character>

Parameter

<Data object> := P\_<Character>

- The definition of field symbols must be based on the data definitions listed above.

| Type                                               | Data type                                                                                                  | Naming conventions |
|----------------------------------------------------|------------------------------------------------------------------------------------------------------------|--------------------|
| Global-Constant                                    | Constant                                                                                                   | GC_<Characters>    |
| Global-Data                                        | Variable                                                                                                   | GV_<Characters>    |
|                                                    | Structure                                                                                                  | GS_<Characters>    |
|                                                    | Table                                                                                                      | GT_<Characters>    |
|                                                    | Reference                                                                                                  | GR_<Characters>    |
| Global-Field-Symbol                                | Variable                                                                                                   | <GV_<Characters>>  |
|                                                    | Structure                                                                                                  | <GS_<Characters>>  |
|                                                    | Table                                                                                                      | <GT_<Characters>>  |
|                                                    | Reference                                                                                                  | <GR_<Characters>>  |
| Local-Constant                                     | Constant                                                                                                   | LC_<Characters>    |
| Local-Data                                         | Variable                                                                                                   | LV_<Characters>    |
|                                                    | Structure                                                                                                  | LS_<Characters>    |
|                                                    | Table                                                                                                      | LT_<Characters>    |
|                                                    | Reference                                                                                                  | LR_<Characters>    |
| Local-Field-Symbol                                 | Variable                                                                                                   | <LV_<Characters>>  |
|                                                    | Structure                                                                                                  | <LS_<Characters>>  |
|                                                    | Table                                                                                                      | <LT_<Characters>>  |
|                                                    | Reference                                                                                                  | <LR_<Characters>>  |
| Local-Statics                                      | Variable                                                                                                   | SV_<Characters>    |
|                                                    | Structure                                                                                                  | SS_<Characters>    |
|                                                    | Table                                                                                                      | ST_<Characters>    |
|                                                    | Reference                                                                                                  | SR_<Characters>    |
| TYPE                                               | Variable                                                                                                   | TV_<Characters>    |
|                                                    | Structure                                                                                                  | TS_<Characters>    |
|                                                    | Table                                                                                                      | TT_<Characters>    |
|                                                    | Reference                                                                                                  | TR_<Characters>    |
| Parameter                                          |                                                                                                            | P_<Characters>     |
| Parameter-Using/Importing                          | Variable                                                                                                   | IV_<Characters>    |
|                                                    | Structure                                                                                                  | IS_<Characters>    |
|                                                    | Table                                                                                                      | IT_<Characters>    |
|                                                    | Reference                                                                                                  | IR_<Characters>    |
| Parameter-Exporting(not allowed for Form routines) | Variable                                                                                                   | EV_<Characters>    |
|                                                    | Structure                                                                                                  | ES_<Characters>    |
|                                                    | Table                                                                                                      | ET_<Characters>    |
|                                                    | Reference                                                                                                  | ER_<Characters>    |
| Parameter-Changing                                 | Variable                                                                                                   | CV_<Characters>    |
|                                                    | Structure                                                                                                  | CS_<Characters>    |
|                                                    | Table                                                                                                      | CT_<Characters>    |
|                                                    | Reference                                                                                                  | CR_<Characters>    |
| Parameter-Returning(not allowed for Form routines) | Variable                                                                                                   | RV_<Characters>    |
|                                                    | Structure                                                                                                  | RS_<Characters>    |
|                                                    | Table                                                                                                      | RT_<Characters>    |
|                                                    | Reference                                                                                                  | RR_<Characters>    |
| Select-option                                      |                                                                                                            | S_<Characters>     |
| Table Parameter                                    | Table – FORBIDDEN (Exception: Remote FUNC)                                                                 | TT_<Characters>    |
| Local Class                                        | definition and implementation                                                                              | LCL_<Characters>   |
| Local Method                                       | <Verb>_<Characters><br><Verb> := <b>get, set, calculate, etc.</b><br><b>Event-Methods:</b> ON_<Characters> | SET_STATUS         |



## 3.5 Patterns

### 3.5.1 EON Header

```
*****
* Brief description of the function:
* (Update of the service package characteristic for equipment in this
* class Sockets dependent on the Property characteristic)
* (Read and/or write access to the data?)
* (Access procedure for modifying access?)
* (Technical dependencies?)
* (Performance requirements?)
* (Sequence control?)
* (Copy of #?)
*
* SAP Release: = XXX X.X
*****
```

### 3.5.2 Change history

```
*****
* Change history:
* Date      User      Reason (Description and CR/ORDER/TICKET ID)
* -----
* DD.MM.YY XXX      XXX
*
*****
```

### 3.5.3 Authority Check

```
INITIALIZATION.
  AUTHORITY-CHECK OBJECT 'S_TCODE'
    ID 'TCD' FIELD gc_tcode.
  IF sy-subrc <> 0.
    MESSAGE a172(00) WITH gc_tcode.
  ENDIF.
```

## 3.6 Modularization of Reports

- Appropriate **Program design** is necessary for faster handling/ easier reading
- Unless previously divided up into modules by means of Includes, the source code **must be broken up into the following sections**. Each section is preceded by comments, and the order of blocks below must be maintained:
- Data definitions should be bundled and placed at the beginning of a section (TOP include, or by form routines/ right after the FORM/ command). Exception is in Inline programming.
  - Declarations
    - Type definitions (TYPES)
    - Global data (DATA)
    - Constants (CONSTANTS)
    - Miscellaneous
    - Selection screen (SELECT-OPTIONS, PARAMETERS, etc.)
  - Events
    - INITIALIZATION

- AT SELECTION SCREEN
- START OF SELECTION (always provide this)
  - Form routines (and modules, as appropriate)
- If the program would be too long or the transparency would be in danger, then divide the code into the following includes:
  - <Report name> - Main Program
  - <Report name>\_TOP - for global declarations
  - <Report name>\_SEL - for the selection screen
  - <Report name>\_DEF - for class definition
  - <Report name>\_IMP - for class implementation
  - <Report name>\_F<digit><digit> - for the form routines
  - <Report name>\_O<digit><digit> - for PBO modules
  - <Report name>\_I<digit><digit> - for PAI modules
- There can be more than one F, O or I include, if it's necessary, but don't overcomplicate.
- INCLUDEs should not be used in more than one program. Use classes instead.

### 3.7 Legal requirement for SAP development and implementation

The majority of SAP systems process data applicable for Internal Control over Financial Reporting. The following legal requirements apply:

- § 239 HGB ["Handelsgesetzbuch"]: German Commercial Code regarding truth and fairness of bookkeeping
- §§145 – 147 AO ["Abgabenordnung"] German fiscal law regarding truth and fairness of bookkeeping
- GoBD ["Grundsätze zur ordnungsgemäßen Führung und Aufbewahrung von Büchern, Aufzeichnungen und Unterlagen in elektronischer Form sowie zum Datenzugriff"] based on a letter from the Federal Ministry of Finance dated 14<sup>th</sup> November 2016

According to these regulations E.ON has to design and operate controls to protect data applicable for Internal Control over Financial Reporting against loss and to assure their integrity and confidentiality. All changes of systems processing such data need to follow a documented procedure ("Verfahrensdokumentation").

### 3.8 Contractual obligations for SAP development and implementation

E.ON Unit – E.ON Business Services (EBS): According to the IT Framework Agreement, section 18.1 "EBS shall provide security for the IT systems in accordance with the rules set forth in the "GP 3-19 E.ON Information Security", in the respective relevant version."

The requirements of the „E.ON Business Directive on GP 3-19\_08 – Rollout, Major Change to and Disposal of Information and Network Systems“ with regard to coding apply.

E.ON Unit – 3<sup>rd</sup> Party Provider: E.ON's General Terms and Conditions of E.ON SE for the Provision of Software Maintenance, the minimum requirement, referring to section "21. Information Security", apply independent from protection level:

- "The supplier undertakes to ensure its organizational and technical IT operations compliant with the international standard ISO/IEC 27001: „Information technology - Security techniques - Information security management systems - Requirements“.



- The supplier undertakes to provide a compliance report confirming that its organizational and technical IT operations are in conformity with the above standards.”